THINKING ABOUT NON-AGENCY RMBS CREDIT

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Agenda

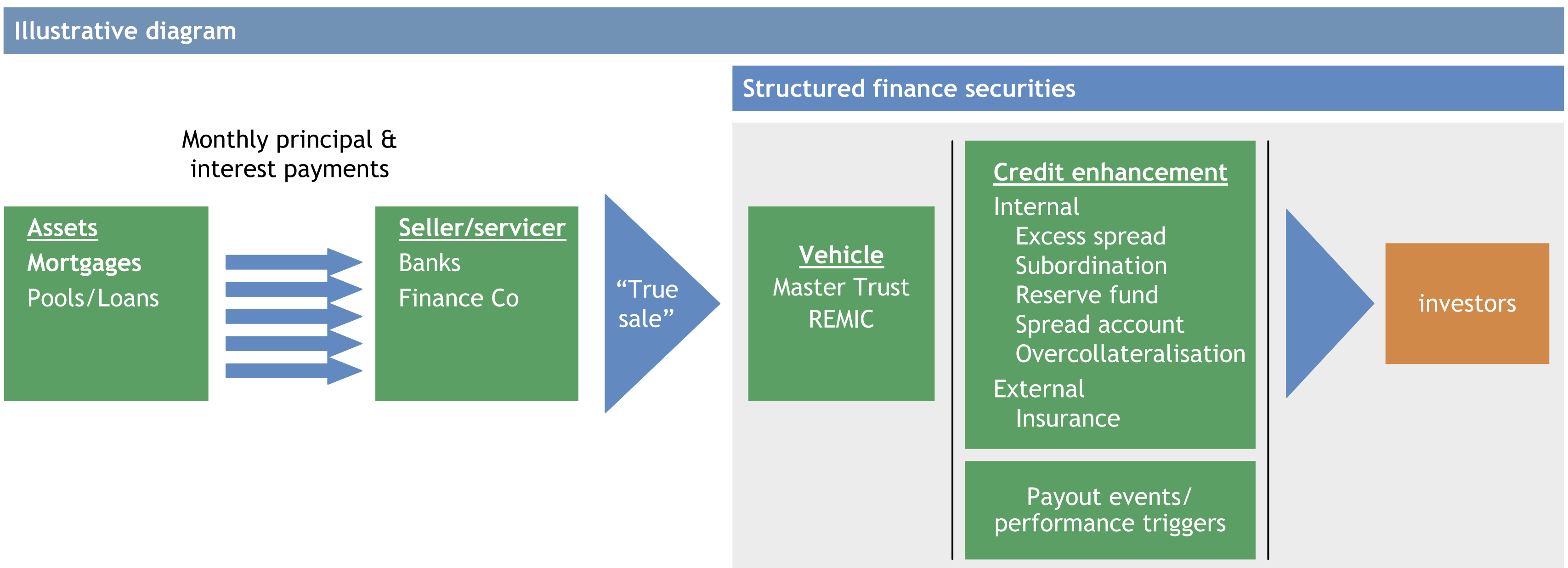
Securitization

Securitization – the process of pooling mortgage loans and converting them into packages of securities with various credit ratings.

- Originators sell whole loans (packages via a flow program) to a Wall Street Dealer or directly into a trust,
- Trust issues rated, registered securities, which the dealer distributes to investors acting as securities underwriter,
- AAA rating is achieved through senior/subordinate structure, whereby a portion of the pool is subordinated to the rest with respect to realized losses,
- Dealer works with investors to structure various cash-flows to meet investor needs and requirements (reverse inquiry),
- Trust receives cash flows from underlying loans and distributes monthly to investors according to distribution rules,
- Dealers provide secondary trading liquidity, valuation and analysis.

Standard securitisation technology

- Represent ownership interest in a pool of assets sold by originators into a special purpose vehicle
  - Are typically secured by homogeneous assets with relatively predictable cash flows
- Repayment of debt is derived from cash flow generated by the underlying assets
- Assets are legally separated from the seller/servicer, limiting investor exposure to the seller/servicer
- Credit enhancements are applied to cash flows to create tailored credit ratings



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A typical prime non-agency CMO structure

Mortgage Pool

Class Size (% of collateral)	**Credit Support** (% of collateral)	Class	
96.50 %	**3.50 %**	AAA	Seniors
1.50 %	**2.00 %**	AA	Investment Grade Subordinates
0.70 %	**1.30 %**	A	Investment Grade Subordinates
0.40 %	**0.90 %**	BBB	Investment Grade Subordinates
0.40 %	**0.50 %**	BB	Non-Investment Grade Subordinates
0.30 %	**0.20 %**	B	Non-Investment Grade Subordinates
0.20 %	**0.00 %**	NR	Non-Investment Grade Subordinates

Source: JP Morgan Securities Inc.

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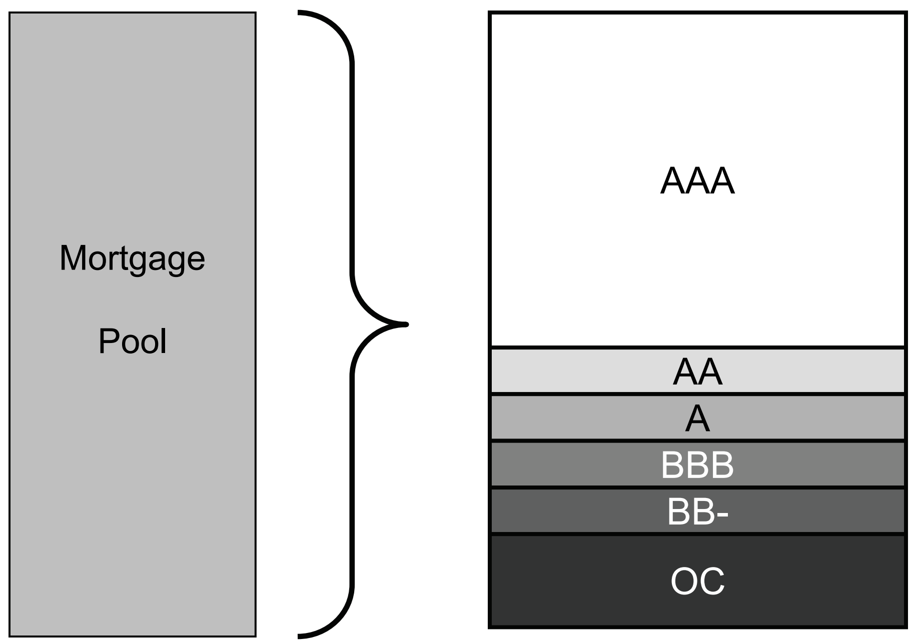
Additional protection: lockout and over-collateralization

- Subordinate bonds are typically **locked out** of principal payments for a number of years (three years on average, but can be affected by delinquency triggers)
  - Senior bonds will receive a disproportionate amount of prepayments
  - Over time, the bond will de-lever as the senior bond pays down faster than subordinates
- Many Alt-A deals featured **excess spread** (interest received from mortgage payments is greater than debt interest payable). This amount is added to a reserve account called **over-collateralization (OC)** that serves as a buffer against losses



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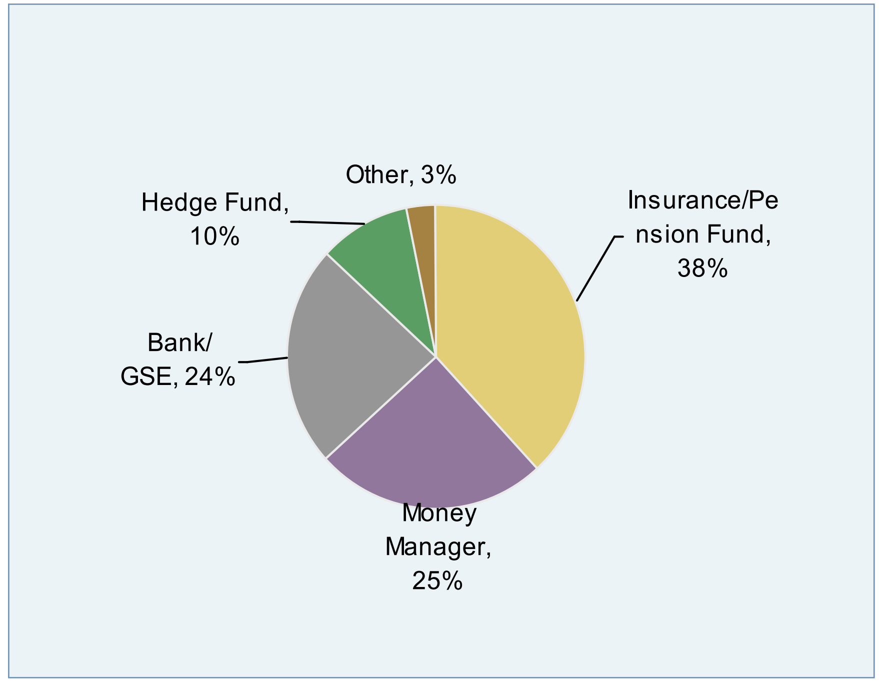
Who invested in Securitized Products?

Typical buyers by bond ratings

Class rating	Typical buyers
AAA	Insurance companies, pension funds, hedge funds, money managers, government sponsored entities and banks
IO	Insurance companies and hedge funds
AA A	Money managers and insurance companies
BBB BBB-	CBO/CDO issuers, insurance companies, pension funds and money managers
BB B NR	B-piece buyers

2005 Participation by Investor Type



Credit support example

Thickness (% of collateral)	**Credit Support** (% of collateral)	
96.50 %	3.50 %	1A1
1.50 %	2.00 %	M1
0.70 %	1.30 %	B1
0.40 %	0.90 %	B2
0.40 %	0.50 %	B3
0.30 %	0.20 %	B4
0.20 %	0.00 %	B5

Credit Waterfall

Source: JP Morgan Securities Inc.

Suppose 1% of the pool defaults next month, resulting in 40bps of losses. Which tranches are written down?

What if losses next month are 150bps?

If lifetime losses are 500bps, which tranches are eventually written down?

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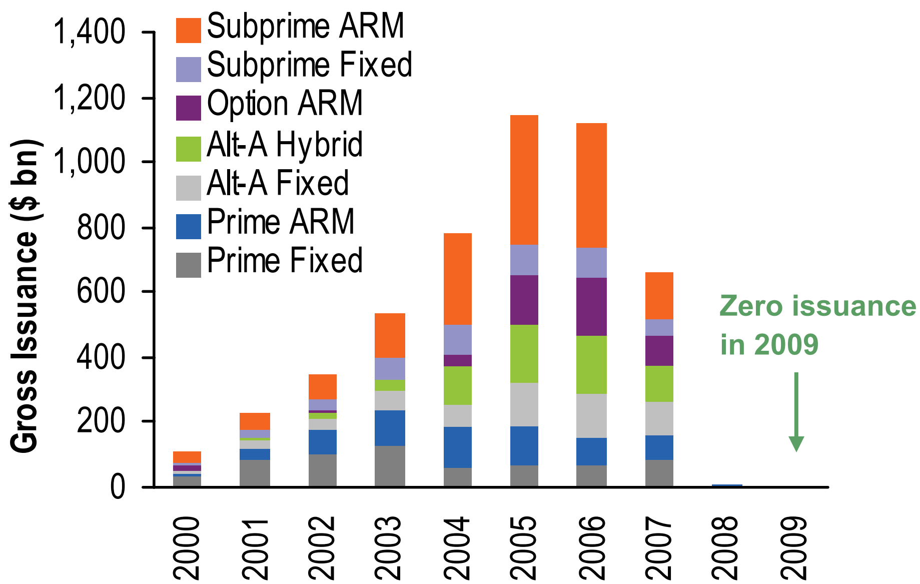
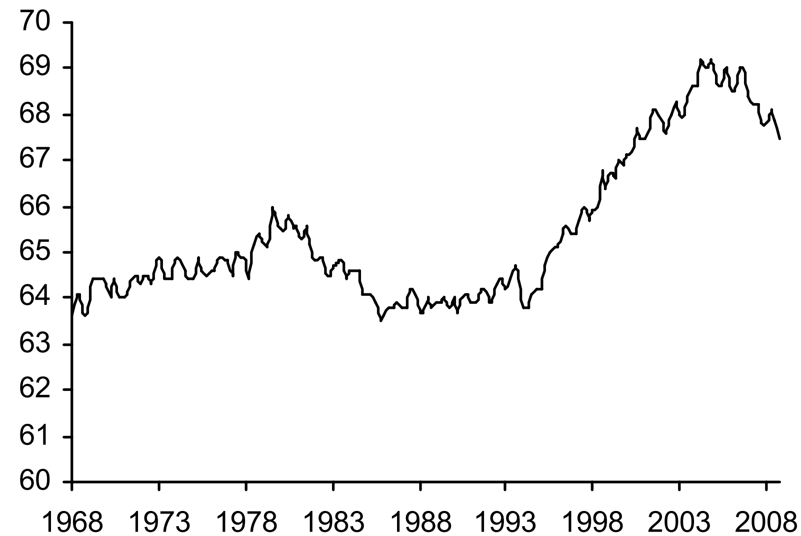
Affordability products exploded during the housing boom as originators scrambled to gain market share

Gross issuance of non-agency RMBS, 2000-present

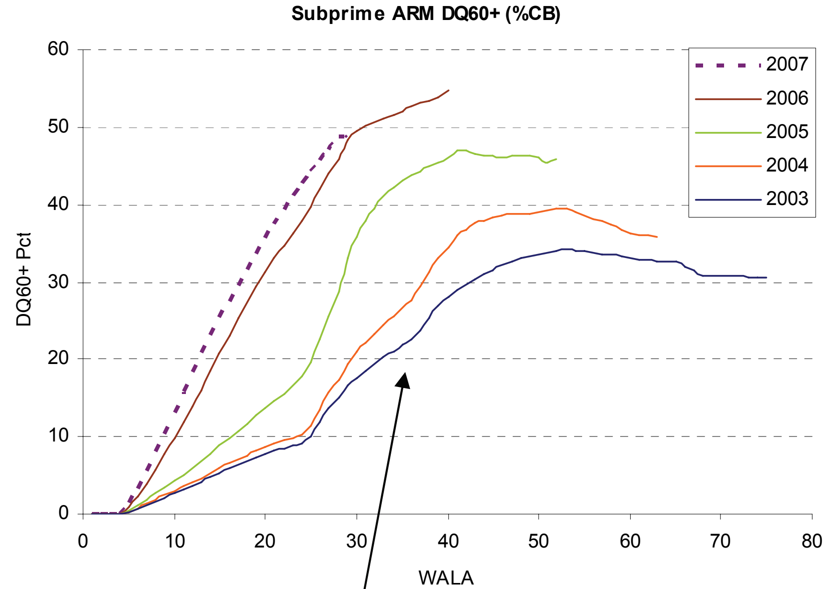


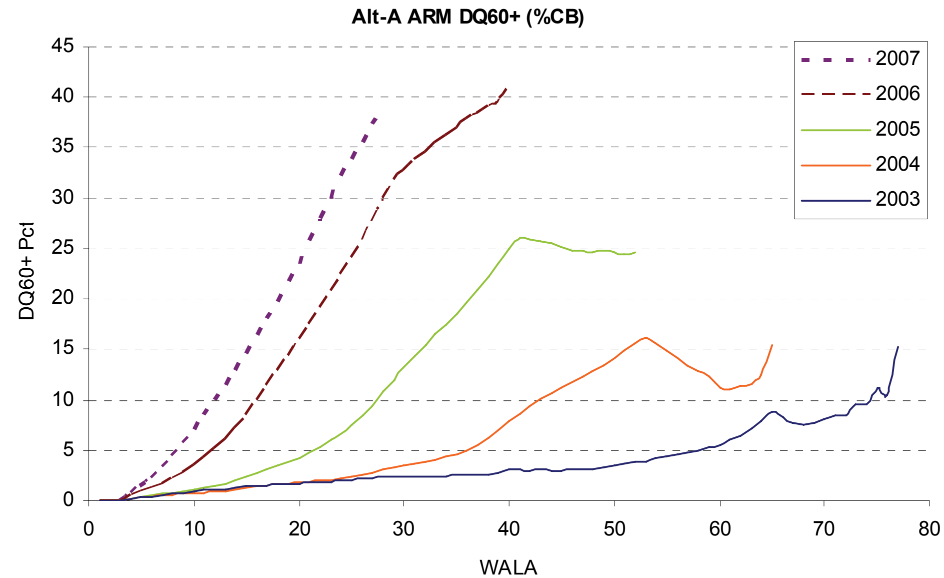
Homeownership rate (%) soared to historic highs



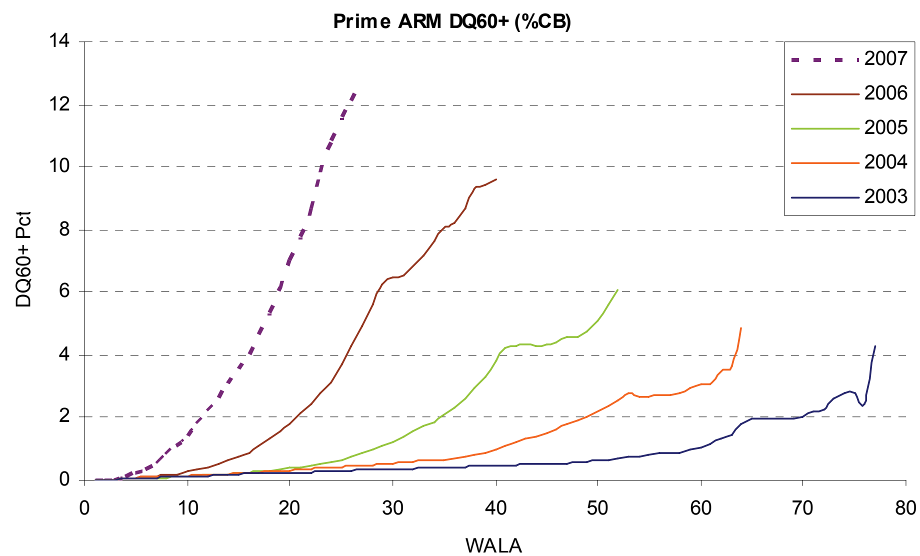
Source: JPMorgan, Loan Performance, US Census

The credit crisis started with delinquencies growing faster than expected…





Underwriting standards expanded each year



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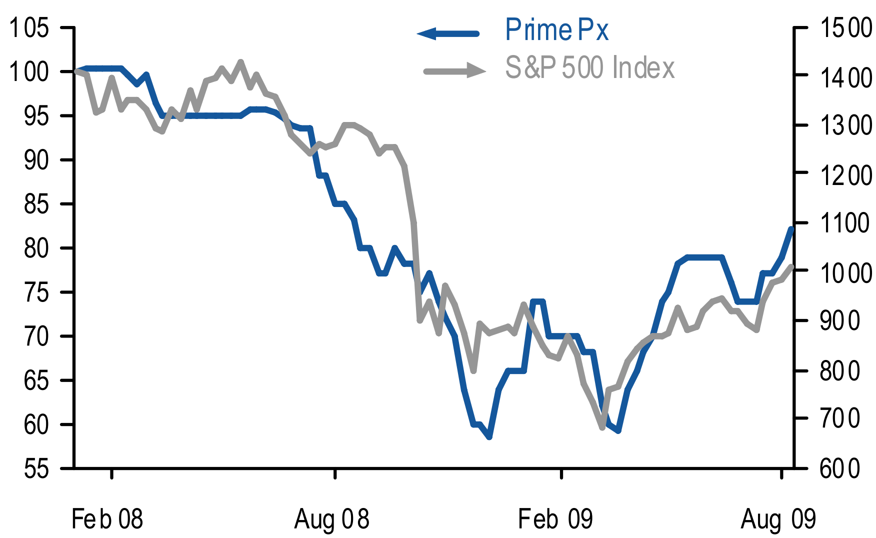
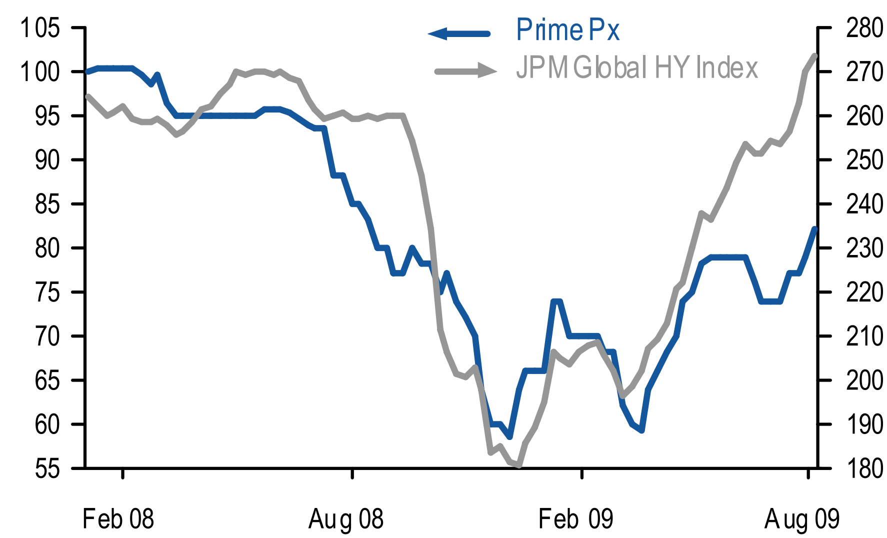
Non-agencies plummeted along with other assets

Prime AAA fixed-rate prices (2006/2007 originations), vs S&P 500 index and vs JPM Global High Yield index

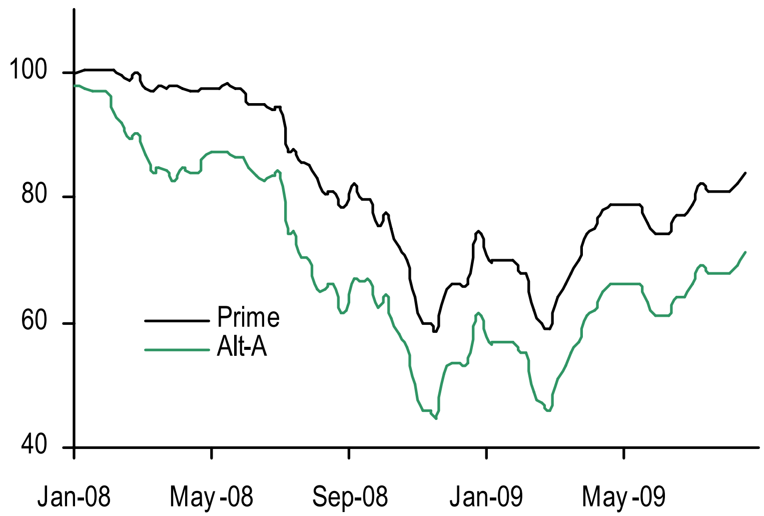




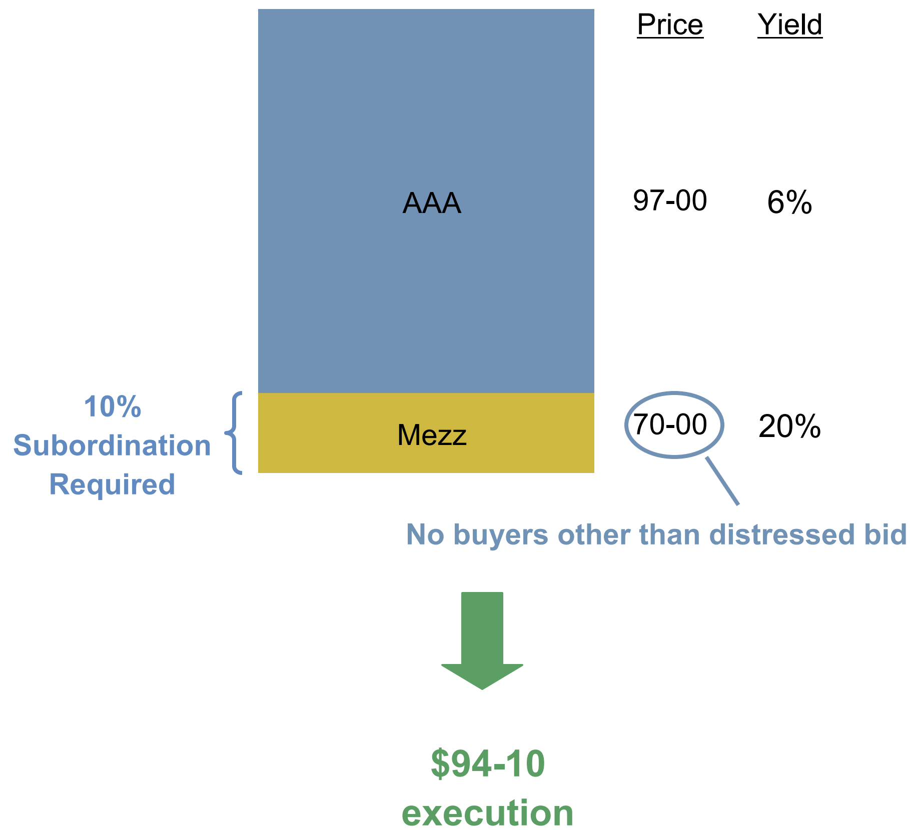
Source: JPMorgan

Issuance stopped in 2008 as execution became prohibitive

Prime and Alt-A prices, Jan 08 - Sep 09



Economics of Securitization
6.5% WAC , $100 origination, FICO 740, LTV 60, Full Doc



Source: JPMorgan

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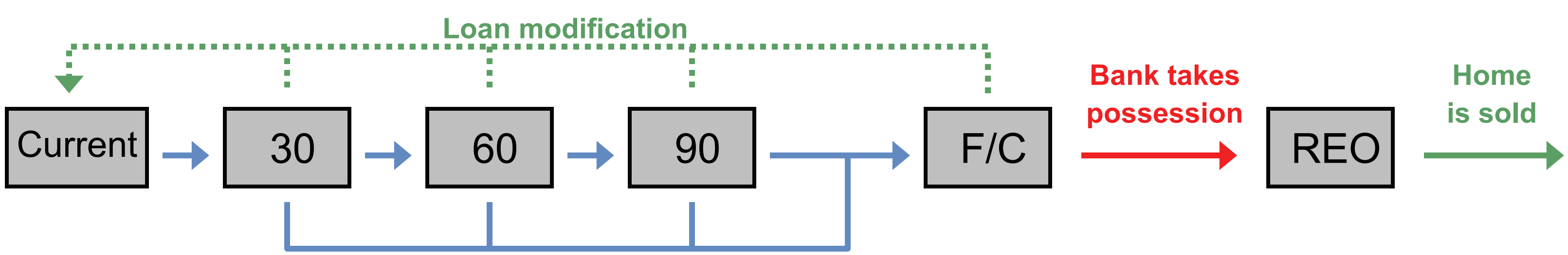
Key Terms

- **Delinquency: 30day, 60day, 90day, Foreclosure, Real Estate Owned (REO)**
  - Borrower has missed (or late) payments
  - Two conventions: MBA and OTS
    - MBA: a borrower is considered 30days delinquent that month if payment is not made by the end of business one day prior to the due date.
    - OTS: a borrower is considered 30days delinquent that month if payment is not made by the end of business of next month's due date.

Essentially, a borrower that misses one payment is current under the OTS method and 1-month delinquent under the MBA method.

Prime tends to use MBA while non-prime tends to use OTS.



Key Terms (continued)

- **Default** – Generally, 120days past due or in foreclosure/REO process
- **Liquidation** – home is sold and proceeds are returned to investors
    - **CDR**: much like CPR, is defined as the annualized default rate.
- **Loss Severity** – Once the property is liquidated, a certain amount of principal is recovered. Often, the percent of balance not recovered is quoted as a "Severity %"
- **Recovery Lag** – Time from default to liquidation
- **Servicer Advancing** – During the recovery process, servicers may advance principal and/or interest payments to the Trust.
- **Voluntary Speed / Constant Repayment Rate (CRR)** – Annualized rate of non-default related prepayments (eg. curtailments, turnover)

Key Terms (continued)

- **LTV** – Loan to value. This is the ratio of loan balance to home price
- **FICO** – A borrower credit score based on a system developed by Fair Isaacs
- **DTI** – Debt to income. Front DTI is the ratio of mortgage payment plus property tax to monthly income
- **60+ DLQ** – Percent of loans in a pool that are 60 days or more delinquent, including loans in foreclosure or are real estate owned
- **Credit Support or Credit Enhancement (C/E)** – Level of insurance against losses provided by subordination. The credit enhancement is the percentage of the deal which is junior to a given bond, and hence will absorb losses first
- **Senior bond** – in a non-agency CMO, a bond that incurs losses only after other bonds in the deal have been written down to zero
- **Mezzanine or subordinated bond** – A bond that incurs losses before senior bonds, thus providing a level of insurance to the senior bond

Putting it all together: example prime bond

JPMMT 2005-A7 1A4

Prime 5/1 SSNR, 6.88% Orig C/E

Hist 3M CRR	18.0
Hist 3M CDR	1.0

60+	10.9
Orig LTV	68.2
Lim Doc	1.5
FICO	742

At 60% Severity

	0 CPR	5 CPR	10 CPR	15 CPR	20 CPR	
0 CDR	**9.86**	**11.36**	**13.17**	**15.22**	**17.52**	*Yield*
	16.6	10.5	7.2	5.3	4.0	*Avg Life*
	0.34%	0.34%	0.34%	0.34%	0.34%	*Proj Cum Loss*
1 CDR	**9.90**	**11.46**	**13.34**	**15.54**	**18.05**	*Yield*
	15.9	10.2	6.9	5.0	3.7	*Avg Life*
	6.50%	4.33%	3.12%	2.40%	1.92%	*Proj Cum Loss*
3 CDR	**9.15**	**10.92**	**12.98**	**15.28**	**17.83**	*Yield*
	13.7	9.2	6.5	4.9	3.8	*Avg Life*
	15.85%	10.77%	7.82%	6.00%	4.81%	*Proj Cum Loss*
5 CDR	**8.21**	**10.05**	**12.20**	**14.59**	**17.23**	*Yield*
	11.8	8.2	6.0	4.6	3.7	*Avg Life*
	22.32%	15.64%	11.62%	9.06%	7.33%	*Proj Cum Loss*
7 CDR	**7.24**	**9.13**	**11.31**	**13.76**	**16.45**	*Yield*
	10.1	7.2	5.5	4.3	3.5	*Avg Life*
	26.82%	19.36%	14.73%	11.68%	9.56%	*Proj Cum Loss*

Risk in non-agencies

For each of the following, pick the category with greater risk of losses to the investor

70% LTV	90% LTV
650 FICO	750 FICO
20% DTI	40% DTI
3% 60+ DLQ	10% 60+ DLQ
10% C/E	20% C/E

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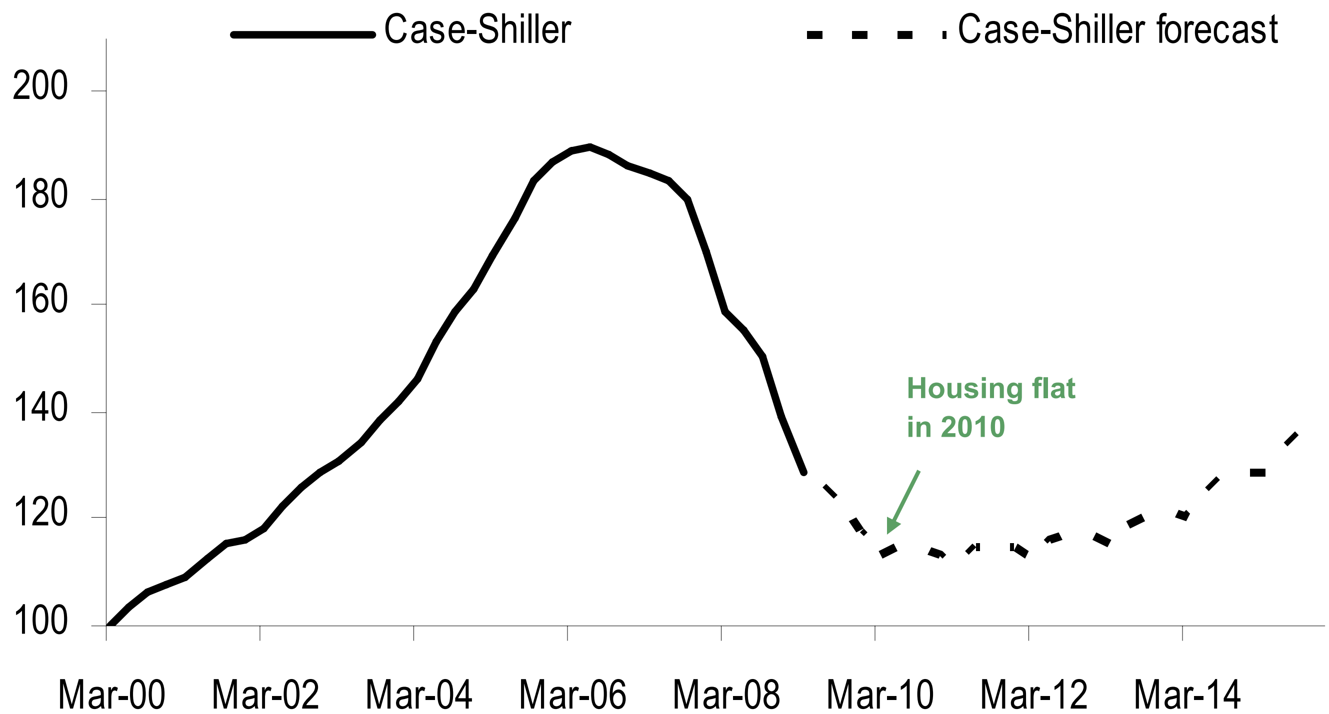
Home price appreciation (depreciation) is a major driver of defaults

Case-Shiller home price index and JPMorgan forecast

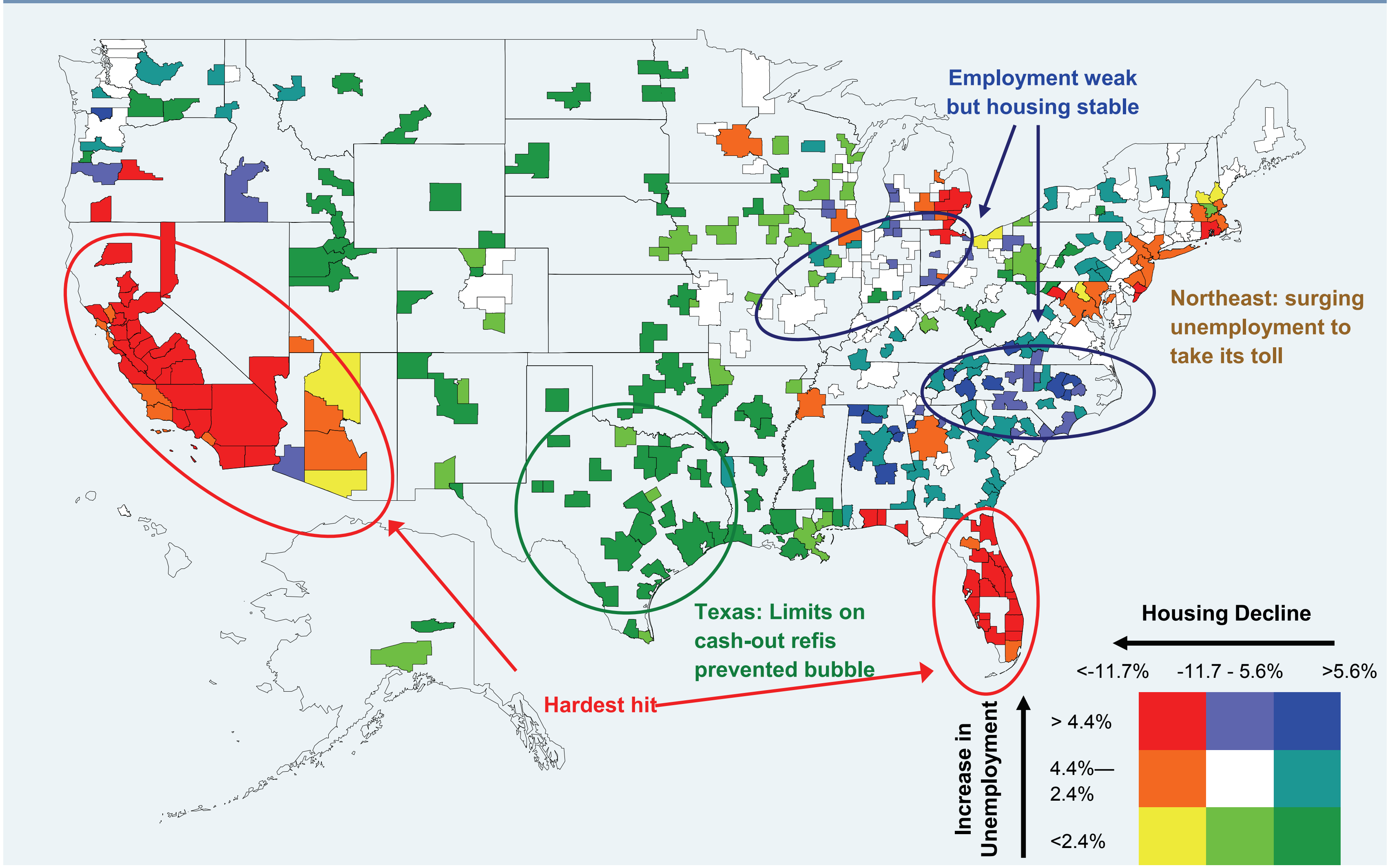


Source: JPMorgan, economy.com

LOSS FUNDAMENTALS

The coasts are hardest hit, with the middle of the country avoiding much of the bubble

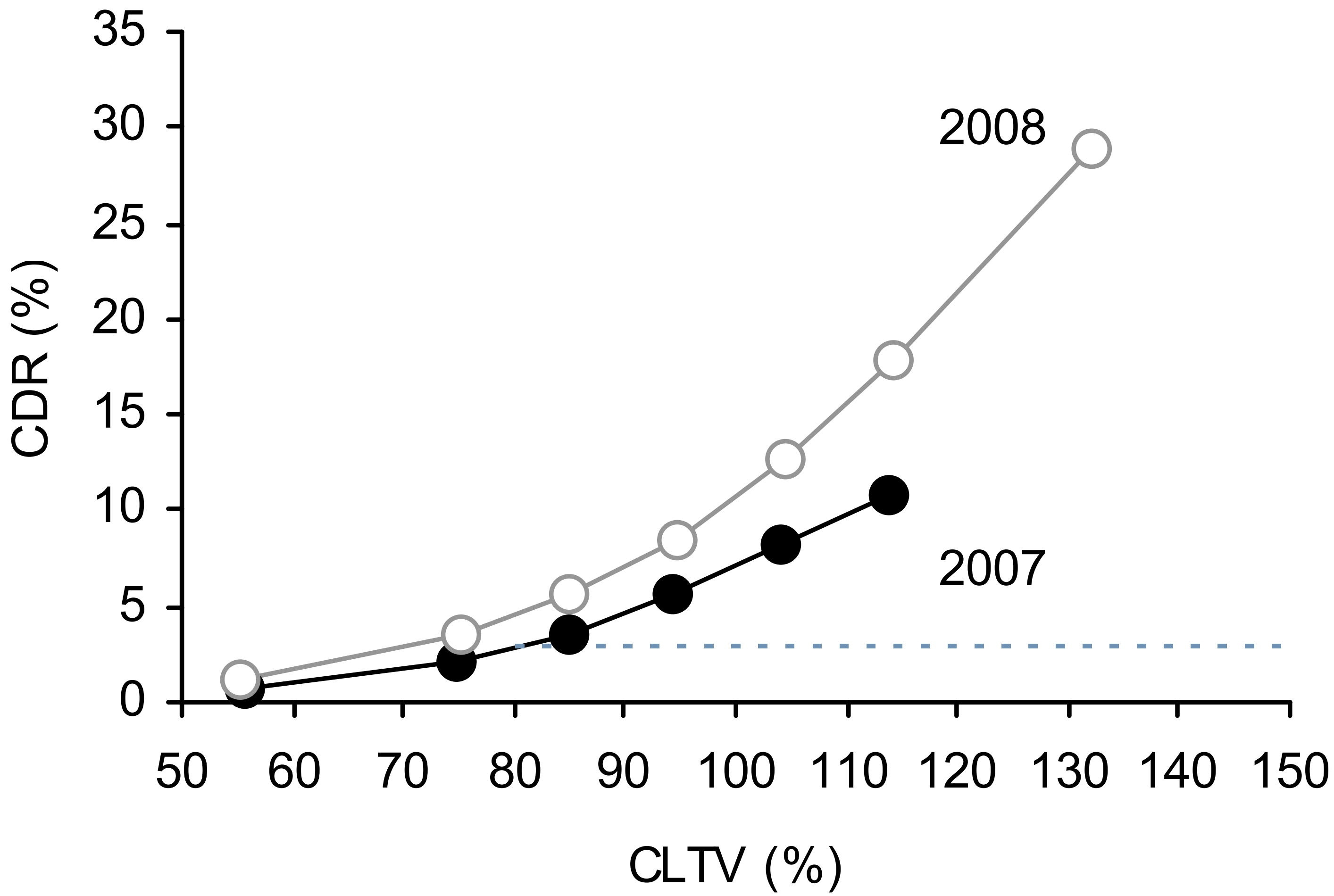
Unemployment pressures home prices, esp. on the coasts



Source: Case-Shiller home prices mid-2006 to 1Q09, OFHEO, Bureau of Labor Statistics Unemployment Rate 06Q2-09Q1

The default optionality curve

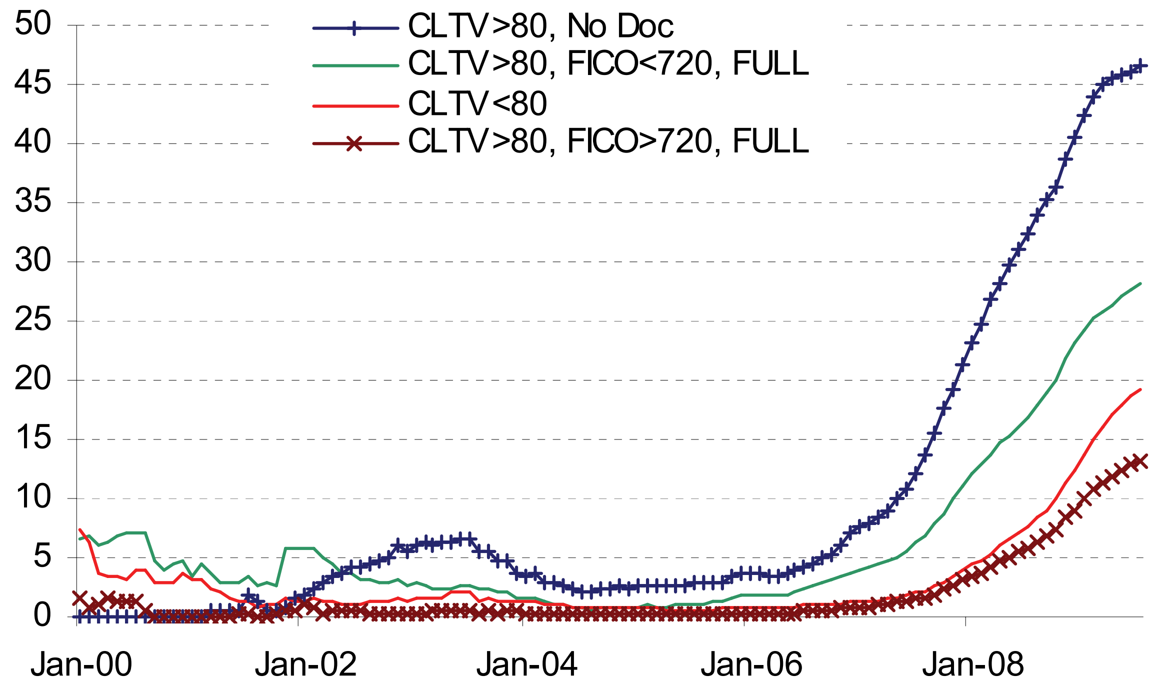
CDR (90-day DQ) of 2005-06 vintage Alt-A fixed, WALAs between 24-36 months, observed in 2007 and 2008



Source: JPMorgan, Loan Performance

Other borrower characteristics driving performance

Alt-A ARMs 60+ day delinquency (%CB) by risk tier



Borrowers with little down payment and weak FICO or no documentation of income

Tier 1 – CLTV<80

Tier 2 – CLTV>80, FICO>720,Full

Tier 3 – CLTV>80, FICO<720, Full

Tier 4 – CLTV>80, Not Full Doc

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Transition model of default

- **The models help project the expected Delinquency Pipeline**
  - Likelihood of defaulting or curing given the current delinquency state is modeled
  - Uses updated information such as pay history to project an event with a short horizon

	Current	30 Day	60+ Day	Fore.	REO	Vol Prepay	Default
Current		x				x	
30 Day	x		x				
60+ Day	x	x	x	x			x
Fore			x		x		x
REO							x

- **The following variables are used to model credit rolls/cures and prepays:**
  - Loan Age
  - Months In State
  - FICO
  - Combined LTV
  - Occupancy
  - Documentation
  - Purpose
  - Prepayment Penalty/Term
  - Payment Shock
  - Number of Missed Payments
  - Past Delinquency
  - Rate Incentive (WAC – Benchmark Rate)
  - Time To Reset (Reset Effects)
  - Loan Size

JPMorgan Loss Expectations

Prime Fixed 30yr

														BondStudio Sev Neg*	
Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	73	44.4	0.3	60.1	739	67.5	67.0	1.42	0.36	28.0	0.3	0.02	2.2	27.8	0.43
2004	61	44.0	2.0	55.3	739	69.7	69.0	2.40	0.51	24.7	0.4	0.05	-17.6	37.8	1.21
2005	48	38.7	0.3	54.0	742	71.3	69.0	4.93	0.69	16.8	1.0	0.16	-29.7	51.7	4.21
2006	36	33.1	0.4	51.0	743	74.2	71.0	7.33	0.66	19.7	2.3	0.35	-32.6	51.8	6.52
2007	26	30.7	0.4	47.8	745	75.7	72.0	6.76	0.77	18.8	1.2	0.25	-29.6	53.3	8.46
Combined	47	37.4	0.6	53.2	742	71.3	69.0	4.89	0.58	21.2	1.1	0.16	-30.8	45.8	4.61

Prime Hybrid ARM

Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	72	43.6	1.5	53.9	733	68.1	67.0	3.51	0.22	27.1	0.9	0.06	-3.3	36.7	0.85
2004	61	51.0	1.7	52.4	735	72.6	69.0	5.41	0.35	24.5	1.7	0.19	-19.0	50.0	2.85
2005	48	50.8	1.9	52.4	741	73.1	70.0	7.93	0.62	17.5	2.5	0.49	-31.7	58.2	7.08
2006	37	51.6	2.5	40.0	740	73.5	70.0	13.25	0.65	16.1	3.8	0.77	-35.1	61.3	10.30
2007	26	52.2	3.1	37.6	741	74.5	70.0	14.78	0.76	14.1	4.3	0.82	-32.6	61.9	11.86
Combined	49	50.3	2.1	48.2	737	71.8	69.0	8.76	0.45	19.7	2.6	0.37	-30.3	54.7	6.59

Alt-A Fixed 30yr

Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	72	35.8	26.7	37.7	711	74.7	73.0	5.88	0.30	12.9	1.1	0.33	-2.5	28.6	1.33
2004	60	28.8	25.2	38.3	710	76.9	74.0	8.92	0.43	10.5	2.0	0.50	-16.9	38.7	3.21
2005	48	27.3	17.7	34.1	713	76.3	72.0	13.58	0.60	7.9	3.4	1.12	-28.1	51.7	10.09
2006	36	22.0	15.2	23.4	706	78.3	74.0	23.17	0.67	7.5	6.4	2.29	-31.0	54.4	18.71
2007	27	25.7	14.5	23.8	713	76.3	73.0	20.69	0.79	7.7	5.4	1.63	-29.0	52.7	18.28
Combined	45	26.4	18.3	30.2	710	76.7	73.0	16.25	0.56	8.6	4.2	1.23	-28.3	48.7	12.21

Alt-A Hybrid ARM

Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	71	41.7	16.1	34.9	709	78.2	74.0	10.65	0.13	12.9	3.5	0.31	-5.3	32.5	1.37
2004	60	40.8	16.4	38.6	709	82.4	76.0	17.63	0.23	10.2	8.5	1.13	-19.1	47.7	5.28
2005	48	38.8	14.4	30.9	713	83.8	75.0	26.24	0.45	5.8	13.0	4.07	-31.7	58.7	17.02
2006	36	42.8	13.3	22.5	711	84.7	76.0	38.32	0.61	4.2	17.0	6.68	-35.4	61.3	28.03
2007	27	50.1	12.7	18.9	718	82.6	76.0	37.15	0.75	4.6	14.8	5.29	-33.0	62.2	30.79
Combined	42	42.5	14.0	26.7	712	83.2	76.0	31.37	0.45	5.6	14.2	4.04	-31.8	58.5	21.87

Source: JPMorgan, Loan Performance

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JPMorgan Expectations (cont)

Option ARM

Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	69	52.2	16.3	29.9	702	71.5	71.0	20.46	0.13	7.0	2.8	0.24	-4.2	41.7	2.17
2004	58	55.1	8.5	24.6	706	73.0	72.0	27.00	0.21	2.7	7.9	0.79	-22.1	54.4	6.24
2005	47	57.2	15.2	16.3	709	77.0	74.0	40.80	0.41	1.3	14.8	3.44	-34.6	65.4	22.18
2006	36	55.6	11.9	9.4	709	79.1	75.0	45.12	0.68	1.1	17.1	5.77	-36.8	67.0	42.00
2007	27	53.9	12.3	11.1	715	78.5	75.0	36.66	0.85	2.4	12.0	3.54	-33.3	66.0	44.19
Combined	38	55.6	12.7	12.3	709	77.5	74.0	41.26	0.54	1.5	15.0	4.02	-35.9	65.7	35.55

Subprime Fixed 30yr

Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	71	24.0	6.7	70.4	641	79.2	79.0	14.60	0.23	8.3	2.8	1.54	-2.8	45.5	5.88
2004	60	28.5	6.0	72.9	641	79.7	79.0	17.30	0.36	6.5	3.4	2.51	-16.2	55.2	12.21
2005	48	21.3	4.9	71.9	639	82.7	81.0	24.66	0.48	3.9	7.0	9.45	-26.7	69.5	29.99
2006	36	16.7	4.5	70.2	635	83.8	82.0	33.39	0.62	3.5	11.0	13.66	-29.5	74.2	47.12
2007	28	13.0	5.0	70.2	622	79.9	79.0	35.10	0.80	2.8	7.5	4.97	-27.4	70.0	47.96
Combined	47	20.7	5.2	71.2	638	81.4	80.0	25.96	0.45	4.7	7.2	6.91	-27.6	65.8	31.21

Subprime Hybrid ARM

Orig Year	WALA	%CA	%Inv	%Full	FICO	Orig CLTV	Orig LTV	60+	Factor	1M CPR	1M CDR	Accum Loss	Est CHPA	Proj Sev	Cum Loss
2003	71	9.4	5.9	68.9	614	83.6	82.0	40.49	0.04	2.5	10.5	2.31	-5.0	36.9	3.86
2004	59	17.5	5.9	65.0	619	85.3	82.0	49.22	0.09	1.9	15.0	3.49	-16.6	49.0	7.92
2005	47	26.3	5.2	57.8	626	86.5	81.0	55.92	0.25	1.0	24.3	9.20	-29.5	61.8	24.97
2006	36	25.7	5.0	55.0	624	87.8	81.0	60.11	0.50	0.9	24.4	13.63	-31.9	68.9	47.39
2007	28	25.3	5.3	58.1	621	85.5	82.0	55.69	0.71	2.2	18.5	8.52	-29.4	69.3	53.37
Combined	41	25.0	5.2	57.2	622	86.2	81.0	57.17	0.27	1.1	22.8	8.20	-29.4	64.8	37.74

** projected losses (including losses that have already occurred)*

Source: JPMorgan, Loan Performance

Agenda

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Government programs

Government Assistance Programs			
Program	**Date Announced**	**Definition**	**Eligibility**
FHA Secure	9/1/07	Offers borrowers caught up in the subprime crisis an option to refinance their current ARM loans to agency insured loans.	They must have made payments on time for at least six months prior to their ARM loan's change date to a higher rate. The effect is to protect these responsible families who were taken in by teaser rates that have since increased (sometimes double or triple) and have caused them to default on their increased payments.
Term Auction Facility (TAF)	12/12/07	The Federal Reserve auctions term funds to depository institutions. The program currently makes available term funds of 28-day or 84-day maturity of up to $125bn per auction conducted every other week or as necessary.	All depository institutions that are eligible to borrow under the primary credit discount window program are eligible to participate in TAF auctions. All advances must be fully collateralized with an appropriate haircut.
Economic Stimulus Act 2008	2/13/08	All eligible individuals were entitled to the basic credit $500. An increase in the conforming loan limit for high-cost areas to $729,750. The total cost of the stimulus is $168bn for 2008.	They need to satisfy at least one of the following two criteria: (1) the sum of earned income (as defined for purposes of the earned income credit) social security benefits must be at least $3,000; or (2) a net income tax liability of at least $1.
GSE Portfolio Caps Raised	3/1/08	OFHEO modifies formula to raise cap on the amount of mortgages the GSEs can retain in their mortgage portfolios.	-
Term Security Lending Facility (TSLF)	3/11/08	Lends up to $200bn of Treasury securities to primary dealers secured for 28 days in exchange for other securities such as agency debt and MBS. This program intends to promote liquidity in the financing markets.	OMO-eligible collateral and investment-grade corporate, municipal, mortgage-backed, and asset-backed securities.
Primary Dealer Credit Facility (PDCF)	3/16/08	An overnight loan facility through the Fed's discount window to give primary dealers access to liquidity in the overnight loan market in order to meet their reserve requirements. The discount window functions as a repo agreement whereby the broker dealer sells a security in exchange for funds, and security in question acts as collateral. The Fed charges an interest rate equivalent to the primary credit rate.	Loans are limited to the amount of margin-adjusted eligible collateral pledged by the dealer and assigned to the New York Fed's account at the clearing bank. Collateral eligible for pledge under the PDCF includes all collateral eligible for pledge in tri-party funding arrangements through the major clearing banks as of 9/12/2008.
Foreclosure Prevention Act 2008	7/30/08	Expands the available tax programs to alleviate the financial pressures on low- and middle-income households. Allows the FHA to insure up to $300bn in new 30yr fixed-rate mortgages for at-risk borrowers. The program cost is funded by fees from Fannie Mae and Freddie Mac, along with fees paid by both lenders and borrowers.	Qualified borrowers must: (1) live in their homes and have loans that were issued between January 2005 and June 2007; (2) have a debt-to-income ratio above 31% of the original loan; (3) can be up to date on their existing mortgage or in default, but must prove that they can no longer keep paying their existing mortgage and attest they are not deliberately defaulting to obtain lower payments. Lenders must agree to write down loan balances to 90% of the homes' current appraised value.
Asset-Backed Commercial Paper Money Market Mutual Fund Liquidity Facility (AMLF)	9/19/08	FRB Boston provides eligible banks non-recourse loans for the purchase of eligible ABCP (asset-backed commercial paper) from a MMMF (money market mutual fund) under certain conditions. The borrower is at no risk of loss on the eligible ABCP unless the ABCP is deemed to be non-conforming. This program intends to generate liquidity in the money markets to prevent default on investors' redemptions.	The ABCP collateral must be: (1) U.S. dollar-denominated; (2) rated at least A-1/P-1/F1; (3) purchased from a MMMF on or after 9/19/2008 at the Fund's acquisition cost as adjusted for amortization of premium or accretion of discount on the ABCP through the date of its purchase. The MMMF must be a fund that qualifies under Rule 2a-7.

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Government programs (continued)

Government Assistance Programs (cont')			
Program	**Date Announced**	**Definition**	**Eligibility**
Hope For Homeowners (H4H)	10/1/08	Provides FHA refinancing for up to $300bn of delinquent fixed rate loans written down by lenders to 90 LTV (or 96.5 if DTI <31). Homeowners participate in equity sharing, whereby they keep more equity the longer they wait to sell or refinance. After 5 years, the FHA shares 50% of equity from sale or refinancing.	Eligible loans must be: (1) owner-occupied properties; (2) originated prior to 1/1/2008; (3) loan limit of $550,440 nationwide; (4) borrowers did not default on the original loan intentionally, cannot have been convicted of fraud in the last 10 years; (5) borrower must be at risk of default but can be current or delinquent; (6) the holder of existing mortgage liens must waive all prepayment penalties and late payment fees.
Troubled Assets Relief Program (TARP)	10/3/08	Allows Treasury to purchase or insure up to $700bn of troubled assets. The program quickly turned into billions of dollars in direct government investments in financial institutions in exchange for partial ownership.	This program demands that companies involved lose some tax benefits, and in many cases incur limits on executive compensation.
Commercial Paper Funding Facility (CPFF)	10/7/08	Provides a liquidity backstop to U.S. issuers of commercial paper through a SPV that purchases 3mo unsecured ABCP directly from eligible issuers. FRBNY provides financing to the SPV, secured by SPV assets and the retention of up-front fees paid by the issuers.	Eligible commercial paper must be: (1) dollar-denominated; (2) rated at least A-1/P-1/F1 by two or more major NRSROs. Limits per issuer: the amount of a single issuer's CP the SPV may own cannot exceed the greatest amount of U.S. dollar-denominated CP the issuer had outstanding between 1/1/2008 and 8/31/2008.
Capital Purchase Program (CPP)	10/13/08	Allows the Treasury to use $250bn of TARP budget to purchase senior preferred shares, which includes warrants for future Treasury purchases of common stock. This program provides capital to viable financial institutions wanting extra capital for stability or lending.	The minimum subscription amount is 1% of institution's risk-weighted assets. The maximum amount is the lesser of $25bn or 3% of risk-weighted assets. The companies involved must lose some tax benefits, and in many cases incur limits on executive compensation. Eligible issuer must be rated at least A-1/P-1/F1 by two or more major NRSROs.
Temporary Liquidity Guarantee Program (TLGP)	10/14/08	FDIC guarantees bank deposits and debts in the event of bankruptcy. There are two components, and institutions can choose to opt out: (1) Debt Guarantee Program guarantees newly issued senior unsecured debt up until 12/31/2012, and (2) Transaction Guarantee Program provides full coverage of non-interest-bearing transaction deposit accounts in FDIC insured institutions until 12/31/2009.	For the Debt Guarantee Program the debt must be issued between 10/14/2008 and 10/31/2009. Eligible entities may issue up to 125% of the senior unsecured debt maturing between 9/30/2008 and 6/30/2009. For the Transaction Guarantee Program the accounts must be above $250,000.
Money Market Investor Funding Facility (MMIFF)	10/21/08	FRBNY provides senior secured funding to SPVs established by the private sector to help restore liquidity to the money markets. Each SPV will purchase eligible money market instruments from eligible investors using financing from the MMIFF and from the issuance of ABCP.	Eligible assets must be: (1) dollar-denominated CDs, bank notes and CP issued by financial institutions; (2) having remaining maturities of 90 days or less; (3) issued by designated financial institutions that have a short-term debt rating of at least A-1/P-1/F1 from two or more major NRSROs. All participating entities are subject to supervisory oversight to prevent rapid growth or excessive risk-taking.
Term Asset-Backed Securities Loan Facility (TALF)	11/25/08	FRBNY lends up to $200bn under TALF 1.0 of non-recourse loans to holders of eligible ABS at a 5-15% haircut. This program is designed to facilitate new issuance of ABS. TALF 2.0 extends up to $1tn and also includes AAA CMBS. Treasury will provide $20bn of credit protection to the Fed in connection with the TALF. Expiration date set at 12/31/2009.	Eligible ABS must have two or more top (e.g., AAA) ratings or, in the case of SBA ABS, must be fully guaranteed by the U.S. government. At the time of a SPV's purchase of a debt instrument issued by a financial institution, the debt instruments of that financial institution may not constitute more than 15% of the assets of the SPV, except during an initial ramp-up period when the concentration limit may be 20%.

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Key programs

Federal Housing Administration

- **Hope for Homeowners (H4H)** – provides FHA refinancing of loans if lender agrees to write down the loan to 90% of the current home price. Only 51 loans closed under the program to date

Treasury (with Troubled Asset Relief Program funds)

- **Making Home Affordable Modification Program (HAMP)** – Provides servicer incentives to modify loans to meet affordability criteria. For borrowers who don't qualify, provides incentives to use short sales or deeds-in-lieu

- **Treasury Public/Private Investment Partnership (PPIP)** – Treasury invests in MBS alongside private investors and provides leverage

Federal Reserve

- **Term Asset-Backed Securities Lending Facility (TALF)** – Provides no-recourse loans to purchase legacy securities. Not currently extended to RMBS

The foreclosure prevention programs: HAMP and H4H

H4H

- Lender writes down loan to 90% LTV. FHA provides government guarantee. Borrower must share any future profits from selling the home with the government
- Epic Fail: Only 51 loans closed to date
- Reasons for failure
  - Second Liens – must be extinguished
  - Early pay defaults (EPD) – if a borrower defaults on the first payment, FHA will not guarantee the loan
  - Complexity – eg. some servicers do not originate loans
  - Capacity and Timing

HAMP

- Servicer reduces interest rate and/or principal to bring borrower's monthly payment below 38% of income. Treasury subsidizes a portion of the reduction and provides cash incentives for each loan modified
- Goal of 500,000 modifications by end of October. This is likely to be reached
- Bulk of servicer focus in terms of modifications

Types of modifications

- **Capitalization of arrearage** – missed payments added to loan balance
- **Term extension** – reamortize the loan to 40-year fixed
- **Creation of IO period** – pay interest only for 5 years
- **Rate reduction** – reduce interest rate for 5 years, then gradually increase
- **Principal forbearance** – convert a portion of principal to a balloon payment and eliminate interest on that part
- **Principal forgiveness** – reduce the amount the borrower owes

"...I COULD'VE HUFFED AND PUFFED, BUT THE PRICE DROPPED SO MUCH I JUST DECIDED TO BUY IT..."

The modification decision

Loan Information	
Orig Loan Bal	400
Orig Home Value	500
Rate	5.00%
Annual Prop Tax	1%
Annual Income	80
Current Conf Rate	5%
NegAm	15%
Housing Environment	
CHPA	-30%
BPO Haircut + costs	20%
DTI	
Current DTI	38%
31% DTI Payment	1.65
New Home Value	350
Unpaid Bal	460
Current LTV	131%

Foreclosure	
Net Sale Proceeds	280
Lender Recovery	**$ 60.87**

Rate Modification		New Balance	460
Target LTV	131%	Rate Reduction	**3.0%**
NPV Loss From		*Pay for Success (1k/yr for 3yrs)*	
New Discount Rate	153	Servicer Incentive	3.6
Principal Forgiveness	-46		
WAC Reduction	29		
Borrower Pays to Maturity		*Borrower Re-defaults*	
New NPV	328	Addtl HPA	-20%
Loss Incurred	132	Net Sale Proceeds	224
		NPV of Sale & Pmts at 7.50%	230
Recovery Value	**$ 71.20**	Price	**$ 50.02**
		Breakeven Re-default	**49%**
Assumed Redefault	53%		
Modification Value	**$ 59.94**		

To mod or not to mod?

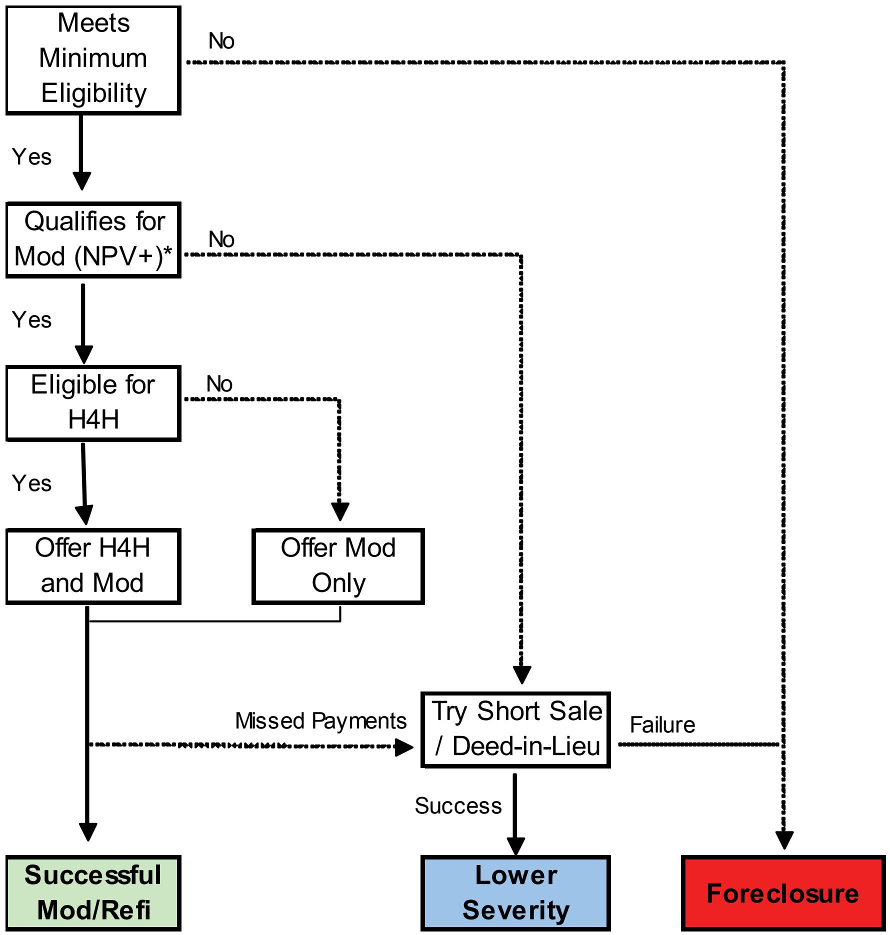
Consider the following scenario.

A mortgage loan made in 2006 for $300,000 is now delinquent. The home can be sold in foreclosure for $140,000 right now or $100,000 in a year. Assuming the borrower is modified and successfully pays to maturity, the lender can recover $200,000 in present value. There is a 50-50 chance the borrower redefaults in a year; otherwise the borrower pays to maturity.

- Is the NPV positive from modification?
- What if the redefault rate were 70%?
- Suppose you forgive principal. The most the lender can now recover is $180,000 but the redefault probability is now 20%. Is the NPV positive?

80% x 180,000 + 20% x 100,000 – 140,000 = $24,000

The complex HAMP decision tree: Modify, H4H, short sale or foreclosure?



If it looks complex, that is because it is. At virtually every node, both the servicer and the borrower are being asked to make decisions and choices that represent significant changes from the choice menu of the pre-crisis regime.

We calculate the NPV as
NPV = (Prob of Redefault × Value from Future Liquidation + Prob of Cure × PV of borrower paying to maturity) − Value of Liquidation

Source: JPMorgan

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Modifications may prevent up to 4 million foreclosures…

Modifications by product type; prime includes conforming loans

Product Type	# Loans	Current LTV	Rate Reduction	Forbearance Amount	% Refinanceable	% Obama Mod	% Forbearance	% Modded to Date	All Mods	Potential Mods
PRIME FIXED	43,018,531	86	1.26	35,010	15%	1%	3%	2%	6%	1.8
PRIME ARM	5,170,288	89	1.20	44,155	4%	1%	6%	3%	9%	0.3
ALT-A FIXED	1,894,295	94	2.86	28,886	7%	3%	11%	3%	18%	0.3
ALT-A ARM	1,154,182	110	2.78	51,004	1%	5%	19%	5%	29%	0.3
Option ARM	1,333,848	113	2.37	76,040	0%	4%	9%	26%	39%	0.2
SUBPRIME FIXED	1,206,674	93	2.38	13,962	8%	8%	20%	9%	36%	0.3
SUBPRIME ARM	2,633,263	110	2.60	32,793	2%	11%	19%	22%	52%	0.8
										4.0

1) **Refinanceable:** These loans are A+ (no missed payments in 12 months), owner occupied, above water, and their refinanced payment is lower than the current payment. We adjust the mortgage rate for loan size (conforming, jumbo conforming and jumbo) and assume that IO/negam borrowers will have to refinance into a fixed-rate fully amortizing loan.

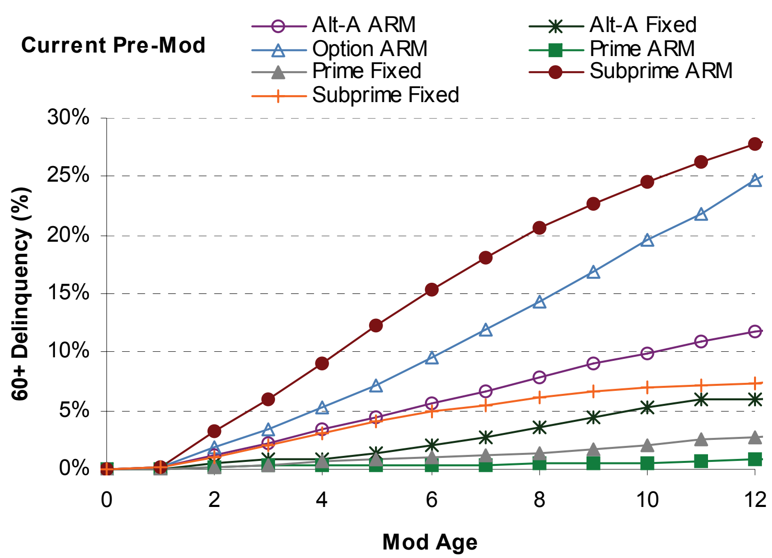
2) **Obama Modifiable:** The loan is not refinanceable, is owner occupied, has balance less than $729,500, and the NPV from modification is positive. We use redefault rates based on prior delinquency status.

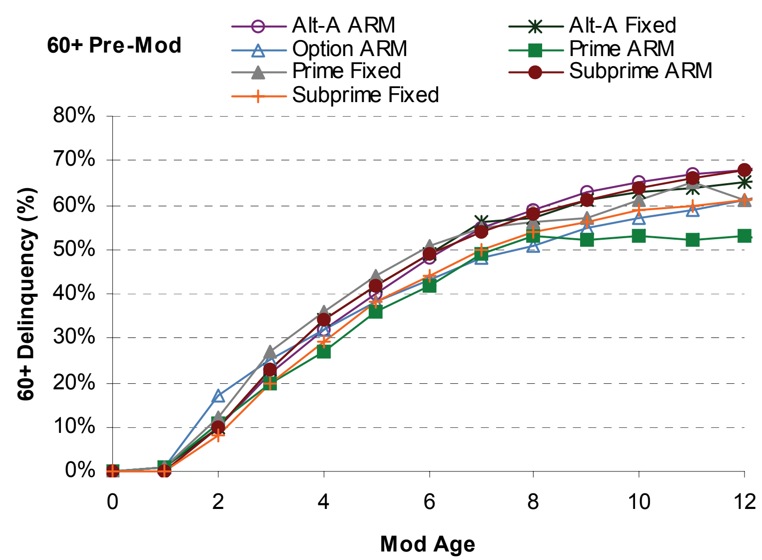
3) **Forbearance Modifiable:** The loan is not refinanceable or Obama modifiable, is owner occupied, and the NPV from forbearance to 95LTV and rate modification to 2% is positive. This represents a streamlined alternative to the Obama plan.

Source: JPMorgan, Loan Performance

Redefault rates for loan modifications are not promising

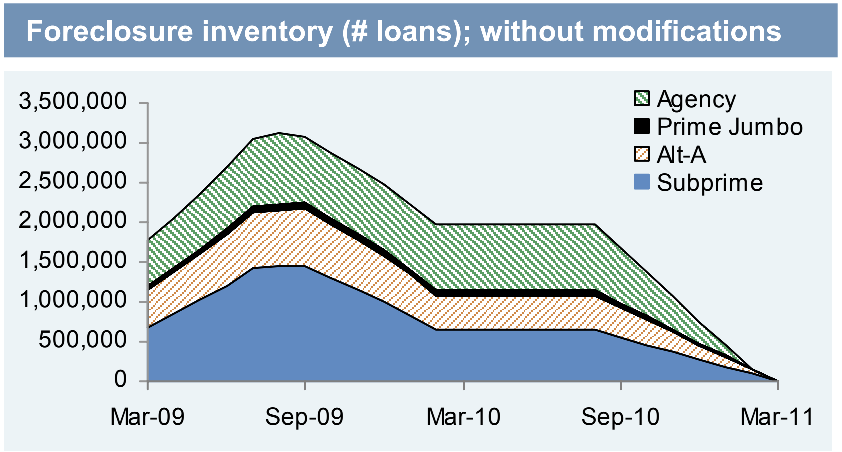
Redefault by product type

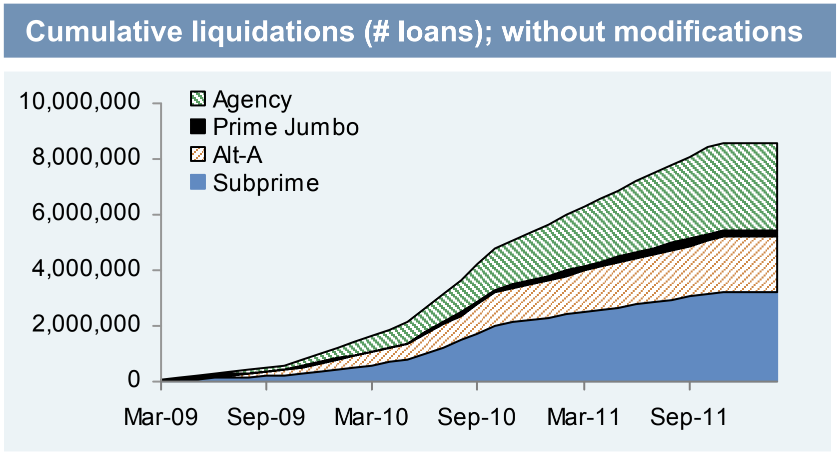


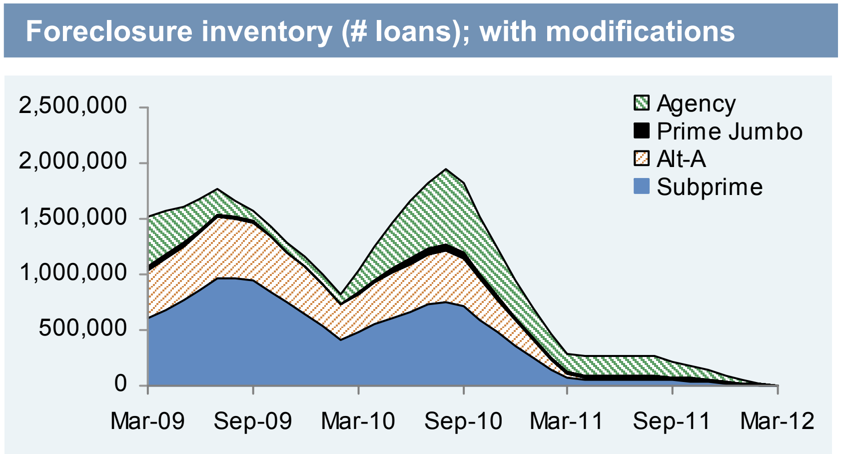


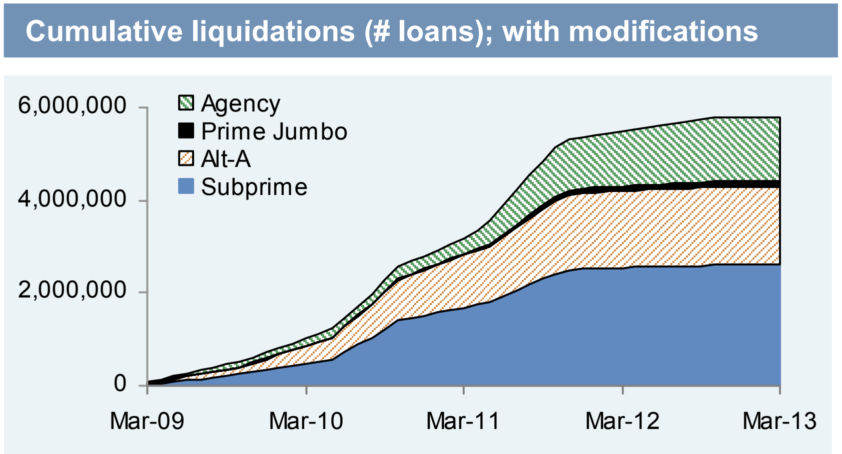
Source: JPMorgan, Loan Performance

Although we project a second wave of defaults, overall foreclosures should be lower due to modifications









Source: JPMorgan, Loan Performance

Agenda

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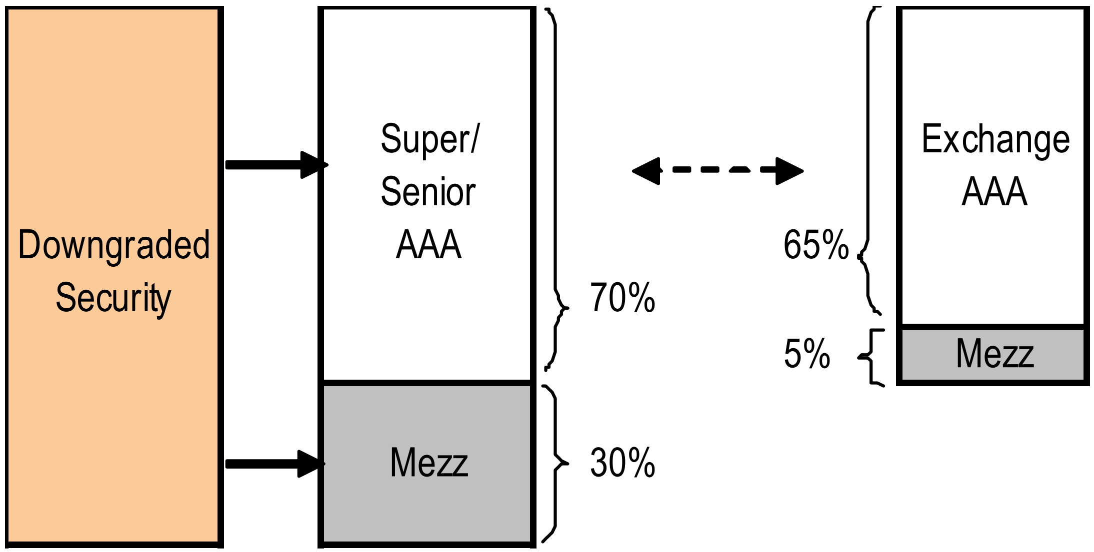
How it works

Example of a Re-REMIC structure



In many deals if the underlying performance is worse than expected, the super/senior Re-REMIC bond can be exchanged for a new bond with greater enhancement

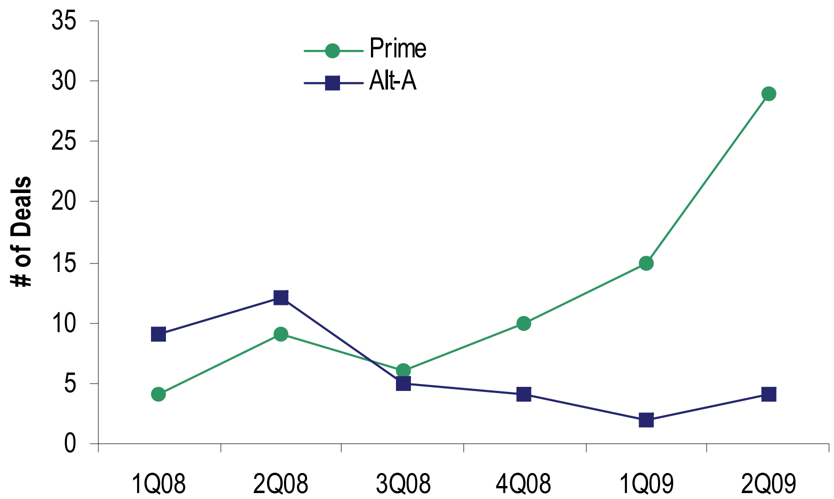
- Cash flows from underlying security are passed through to the Re-REMIC structure
- Support bond is locked out from principal payments until the super/senior is paid off
- Any writedowns to the underlying are first absorbed by the subordinated security

- In a Z structure, the support bond accrues interest but receives no payments until the super/senior has paid off

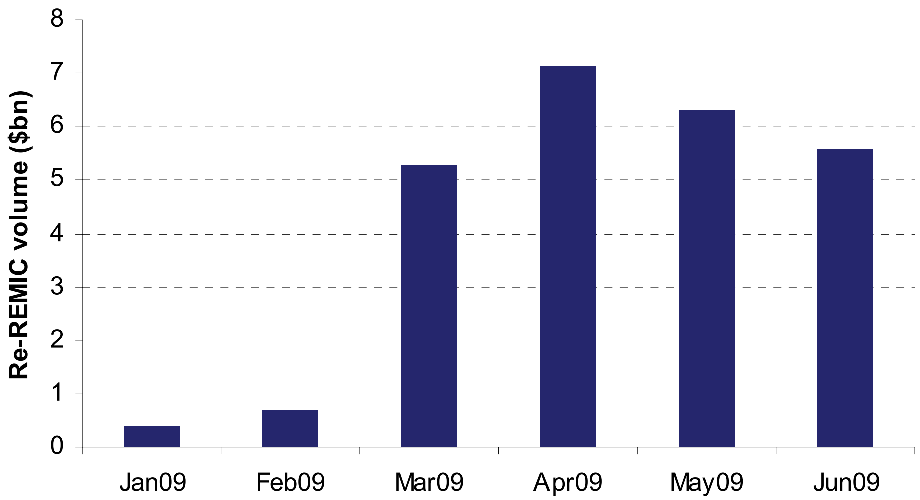
Source: JPMorgan

Prime re-securitizations have surged in 2009

Re-REMIC deals by quarter



1H09 volume of activity



Source: JPMorgan, Bloomberg

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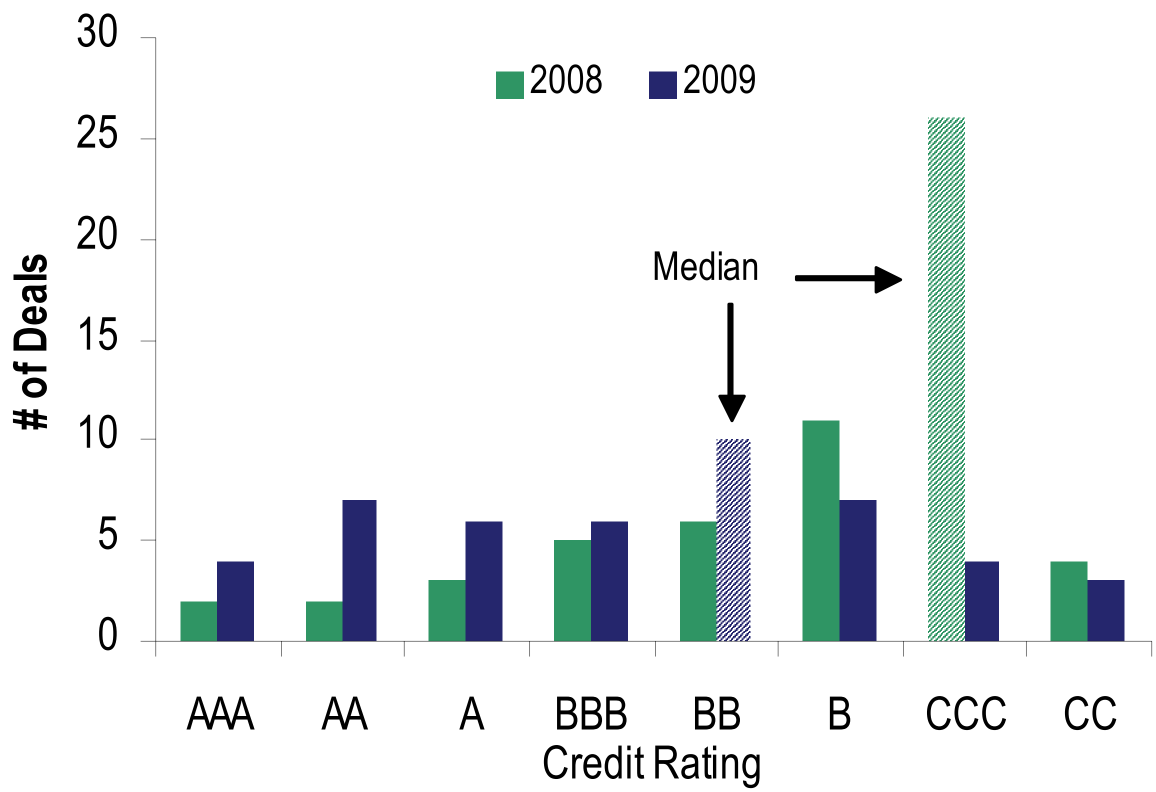
Underlying bonds have been at risk of downgrades and writedowns

Current credit rating of Re-REMIC underlying bonds by year of Re-REMIC issuance



Source: JPMorgan, Bloomberg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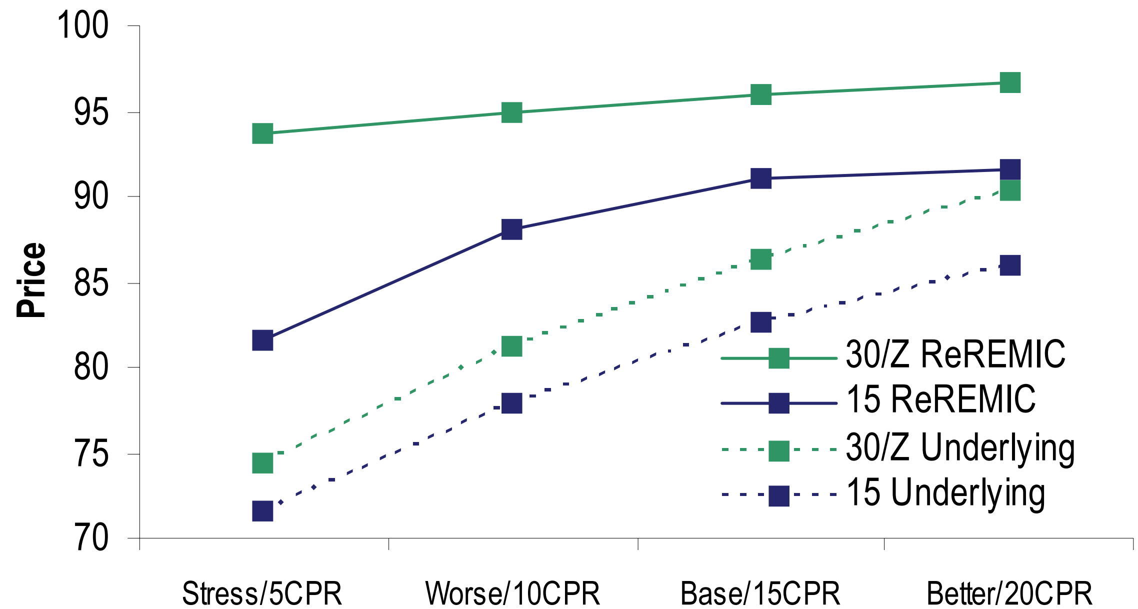
Highly enhanced Re-REMIC super/seniors provide excellent downside protection but cap the upside

Price by scenario and structure for Re-REMIC super/senior and underlying



Assumptions

Bond	Mezz Type	C/E	60+	3M CPR	Base Default Scenario	Stress Default Scenario
JPMRR 2009-5 1A1	SEQ	32	16.2	14.6	8CDR 24mo / 5CDR; 15CPR	11.2CDR 24mo / 7CDR; 5CPR
BCAP 2009-RR2 A1	Z	33	13.6	17.0	6CDR 24mo / 4CDR; 15 CPR	8.4CDR 24mo / 5.6CDR; 5 CPR
CSMC 2009-7R 1A1	SEQ	15	11.2	22.6	5CDR 24mo / 4CDR; 15 CPR	7CDR 24mo / 5.6CDR; 5 CPR
JPMRR 2009-6 1A1	Z	15	8.5	20.8	4CDR 24mo / 3CDR; 15 CPR	5.6CDR 24mo / 4.2CDR; 5 CPR

Source: JPMorgan

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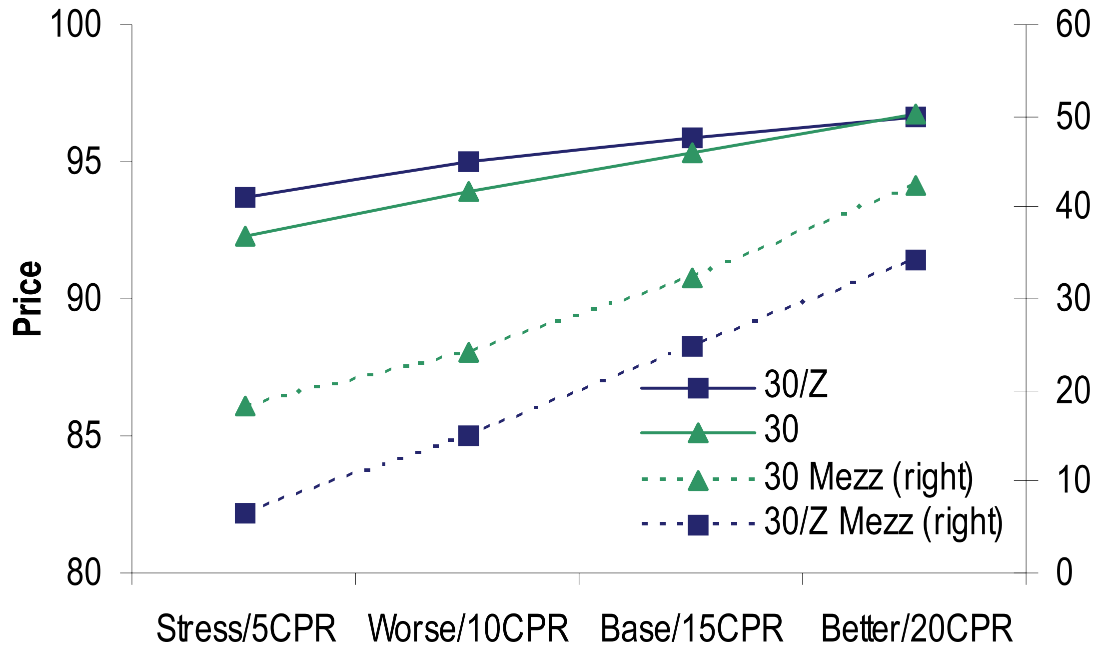
Mezzanine investors seeking to benefit from improving economy must weigh downside risk

Price by scenario and structure for Re-REMICs super/senior and mezz



Typical Yields	
Super/Senior	7%
Mezz	20%

Source: JPMorgan

Analyzing Re-secs

- Between the following, what type of bond is better for a bank or insurance company?

  15% C/E Super/Senior   30% C/E Super/Senior

- How about these two?

  30% C/E Super/Senior   30% C/E Mezzanine

- What about for a hedge fund?

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